Hieu Phan

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EDUCATION

- **2020 - 2023**: Doctor of Philosophy, Thesis Title: "Knowledge Distillation and Continual Learning for Optimized Deep Neural Networks" - University of Wollongong.
- **2017-2019**: Bachelor of Computer Science - GPA 9/10 - University of Wollongong.

ACHIEVEMENTS

- **2023**: $40k research grant [1*] on tumor segmentation from Channel 7 Children's Research Foundation as *Chief Investigator* (media link).
- **2022**: $50k research grant [4*] on scene segmentation for road conditioning assessment from QLD Department of Transport and Main Roads as Co-Investigator.
- **2021-2023**: University Postgraduate Award - Ph.D Scholarship from University of Wollongong.
- **2020**: Best research paper award in IEEE SmartIoT Conference.
- **2017-2019**: Dean's Merit Award (annually for the top 5% in the faculty) - University of Wollongong.

AI/ML EXPERTISE

- Generative AI [2], multi-modal model [1], [3], [7], prompt tuning [4], and explainable AI [13], [3].
- Vision Transformer [6], knowledge distillation for continual learning [5], and model compression [8],[9].
- Natural language understanding [11].
- Applied AI: tumour segmentation, medical image synthesis, and diagnosis [1*,2*,3*,5*].

PROFESSIONAL EXPERIENCE

Google — Sydney
Research Scientist — *Oct-Now*

- Research on multimodal LLMs with enhanced visual perceptions.

Australian Institute for Machine Learning, Adelaide University — Adelaide
Postdoctoral Research Fellow — *16 Nov 2022 - Aug 2025*

- Develop state-of-the-art vision-language models, outperforming radiologists on X-ray disease classification by 6%.
  Boost 30% radiologist report writing time - reduce $3.8K daily cost of radiologists.
  Published at **CVPR2024** [1] (top-1 global AI conference) with 70-star open-source model on Github.
- Develop and deliver tumor segmentation pipeline for patient treatment planning at South Australian hospitals.
  Won three nationally competitive research grants [1*,2*,3*] from clinical research centers, totaling $127k.
- Develop LLM with prompt engineering to combat hallucinations for trustworthy medical applications. Boost disease diagnosis accuracy of LLMs by 5%.
  Collaborate with clinicians to define machine learning scopes. Work with medical professionals to collect, annotate data, and evaluate LLM hallucinations for high-stake disease diagnosis.
- Develop generative models to generate CT images from MRI, avoiding patients' radioactive exposures.
  Minimized harmful CT scans for 72% of pediatric patient intakes.
  Publish advanced generative models at MICCAI2024 [2], and MICCAI2023 [7], top conference in medical imaging (Github repo here).
- Mentorship and leadership: Supervise 2 PhD and 3 postgraduate students on deep learning research. Publish a Vision Transformer model at top IJCV journal (Impact Factor 11.6) [6] and explainable AI at CVPR2024 [13].

University of Wollongong — Wollongong
Machine Learning Engineer — *Nov 2021 - Nov 2022*

- Develop machine learning models for automatic bird counting, fire detection and fire risk assessment models for the AI-IoT environmental management system.
- Deploy AI models on edge using Azure IoT Edge.

- Collaborate with ecologists in Vietnam's Tram Chim National Park to design and deploy a scalable software pipeline tailored for field use.
- Continuous model development: monitored and retrained models to adapt to evolving environmental data, ensuring ongoing accuracy and relevance.
- High-impact recognition: receive endorsement from Australian ambassador and Vietnamese government for its impact in protecting forest and biodiversity: media link, project website.

University of Wollongong Wollongong
AI/ML Research Assistant *Feb 2022 - Nov 2022*
- Develop detection models to assess road conditioning for Department of Transport and Main Roads.
- Build a mapping of 10 road defects and 11 road types on 4,109 km of state-strategic road.
- Awarded a grant [4*] worth $50k from Department of Transport and Main Roads, Queensland, Australia.

ASX Sydney
Data Scientist *Sep 2019 - Feb 2020*
- Develop computer vision model to extract texts and tables from PDF files.
- Build text mining tools to extract entities and their relationship in PDF files.
- Reduced annual data entry costs by $40k, demonstrating measurable business impact through automation.

AI/ML RESEARCH PROJECTS

Multi-modal LLM
Postdoc Research Scientist, AIML, University of Adelaide *Aug 2023-Now*
- Develop vision LLM for high-stake disease diagnosis. Publish at top-tier global conference CVPR2024 [1]
- Work with clinicians, and customers to collect, and annotate data. Design end-to-end pipeline for training and evaluating LLM and vision-language models; align AI solutions with business needs.
- Enhance LLM's reasoning for clinical diagnosis using retrieval-augmented generation (RAG) and chain-of-thought prompting. Publish at **EMNLP**, US, 2024 [4] (top-2 conference in NLP).

Generative model
Research Scientist, AIML, University of Adelaide *Jan 2023- Now*
- Develop Transformer-based generative AI for image generation.
- Resolve structural deformation of current generative models by integrating anatomical priors. Publish MICCAI 2024 (Top 11% - early accept) [2], and MICCAI 2023 [7].

Knowledge distillation for continuous training and model compression
Ph.D, University of Wollongong *Nov 2021-Nov 2022*
- Develop advanced continual learning frameworks to enable continous update and prevent forgetting. Publish CVPR 2022 [5].
- Develop compression algorithms for deployable segmentation models in biosecurity scanning applications. Publish Neurocomputing 2022 [10].

AI-IoT for smart appliances
IoT, ML Engineer *Aug 2020 - Feb 2021*
- Develop an AI-IoT system for smart appliances. Develop ML models classify signals to optimize energy savings and understand consumers' behaviors.
- Deploy models on smart IoT sensors for monitoring energy loads.
  My system achieves Best Paper Award at an international conference, IEEE SmartIoT.

Aspect-based sentiment analysis on product reviews
B.CompSc, University of Wollongong *Jun 2019 - Dec 2019*
- Discover the lack of syntactical understanding of language models, and develop syntax-enhanced LMs. Publish **ACL** 2020 [11], top-1 global conference in NLP; attract 200+ citations (link), showing NLP industry and research impacts.

GRANTS

[1*] Channel 7 Children's Research Foundation, "AI-assisted Contouring of Sarcomas to Improve Safety and Efficiency of Proton Therapy", **Chief Investigator**, $39,570, 2023.

[2*] NeuroSurgical Research Foundation, "Paediatric Radiation Therapy with AI-based Segmentation for Enhanced Treatment Planning", *Associate Investigator*, $47,790, 2023.

[3*] ANZSA, "Improving the Radiological Evaluation of Bone and Soft Tissue Sarcomas with Advanced Diffusion Imaging and Artificial Intelligence", *Associate Investigator*, $40,000, 2023.

[4*] TMR Spatial Labs Program, "Developing Deep Learning Tools for Road Information Extraction from Georeferenced High-resolution Imagery", *Co-Investigator*, $50,000, 2022.

Under review:

[5*] Bloom Research Program, "BRinging Artificial Intelligence iNto pediatric oncology: the myBRAIN translational trial project", *Co-Investigator*, $2M, 2024.

PUBLICATIONS

[1] **V. M. H. Phan**, et al., "Decomposing Disease Descriptions for Enhanced Pathology Detection: A Multi-Aspect Vision-Language Pre-training Framework," *IEEE Conference on Computer Vision and Pattern Recognition*, 2024, Accepted (Top-1 conference in AI, H-index=422); Presentation link.

[2] **V. M. H. Phan**, et al., "Structural Attention: Rethinking Transformer for Unpaired Medical Image Synthesis", *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Early Accept (Top 11%), 2024. Paper link

[3] F. C. Townim, **V. M. H. Phan**, et al., "AdaCBM: An Adaptive Concept Bottleneck Model for Explainable and Accurate Diagnosis", *International Conference on Medical Image Computing and Computer-Assisted Intervention*, 2024.

[4] T. D. Nguyen, T. T. Huynh, **V. M. H. Phan**, et al., "CARER - ClinicAl Reasoning-Enhanced Representation for Temporal Health Risk Prediction", *EMNLP*, 2024 (Top 2, conference in NLP, H-index=193). Paper link.

[5] **M. H. Phan**, T. Ta, S. L. Phung, L. Tran-Thanh, A. Bouzerdoum, "Class Similarity Weighted Knowledge Distillation for Continual Semantic Segmentation," *IEEE Conference on Computer Vision and Pattern Recognition*, 2022 (H-index=422).

[6] B. Zhang*, L. Liu*, **M. H. Phan***, et al., "SegViTv2: Exploring Efficient and Continual Semantic Segmentation with Plain Vision Transformers", *International Journal of Computer Vision*, 2023 (IF=19.5). *Equal contribution

[7] **M. H. Phan**, et al., "Structure-Preserving Synthesis: MaskGAN for Unpaired MR-CT Translation", *International Conference on Medical Image Computing and Computer-Assisted Intervention*, 2023 (H-index=89).

[8] J. Yuan*, **M. H. Phan***, et al., "FAKD: Feature Augmented Knowledge Distillation for Semantic Segmentation", *IEEE Winter Application on Applications of Computer Vision*, 2024 (H-index=95). *Equal contribution.

[9] L. Liu, Z. Wang, **M. H. Phan**, et al., "BPKD: Boundary Privileged Knowledge Distillation For Semantic Segmentation", *IEEE Winter Application on Applications of Computer Vision*, 2023 (H-index=95).

[10] **M. H. Phan**, T. S. L. Phung, K. Luu, A. Bouzerdoum, "Efficient Hyperspectral Image Segmentation for Biosecurity Scanning using Knowledge Distillation from Multi-head Teacher," *Neurocomputing*, 2022 (IF=6.0).

[11] **M. H. Phan**, P. Ogunbona, "Modelling context and syntactical features for aspect-based sentiment analysis," in *Association for Computational Linguistics*, 2020 (Top-1 conference in NLP, H-index=215). Presentation link.

[12] V. K. Nguyen, **M. H. Phan**, "A hybrid approach for intrusive appliance load monitoring in smart home", in IEEE International Conference on Smart Internet of Things (SmartIoT), 2020, *Best Paper Award*.

[13] F. C. Townim, K. Liao, **V. M. H. Phan**, et al., "CAPE: CAM as a Probabilistic Ensemble for Enhanced DNN Interpretation," *IEEE Conference on Computer Vision and Pattern Recognition*, 2024, (Top-1 conference in AI, H-index=422).

REFEREES

- Prof. Son Lam Phung. Email: phung@uow.edu.au.
- Dr Yuankai Qi. Email: yuankai.qi@mq.edu.au.